

Network Effects of Oil Price Shocks on Inflation: A DAG-SVAR Framework for Turkey

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Abstract

We study how Brent oil price shocks transmit to Turkish consumer prices using a two-stage DAG–SVAR framework. Standard structural VARs that target headline CPI typically deliver imprecise pass-through estimates because the headline index aggregates oil-sensitive items together with components (rent, communication, education, etc.) that do not respond to oil shocks. We address this by using the PC causal-discovery algorithms to identify the CPI items that is affected of Brent Oil prices. We then run the SVAR with this DAG-selected basket. The procedure isolates 50 five-digit CPI items and produces a tighter, more clearly identified pass-through coefficient than the headline-CPI benchmark. We also implement VAR-LiNGAM as a robustness check; while non-Gaussianity of residuals is strongly supported by the data, VAR-LiNGAM at the two-digit level suffers from the dimensionality of the implied basket and does not improve precision in our setting.

1 Introduction

Oil prices are one of the most studied drivers of global inflation. The post-COVID supply disruptions, the 2022 Russia–Ukraine episode, and the recurrent tensions involving Iran in 2024–2026 have all renewed central-bank interest in measuring how, and how fast, an international oil shock propagates into domestic consumer prices. For an oil-importing emerging economy such as Turkey, where energy enters production through multiple channels (transport fuel, fertilisers, plastics, electricity generation), this question is not academic: the size of the pass-through coefficient is a direct input to monetary-policy reaction functions.

The standard tool to measure oil pass-through is a structural vector autoregression (SVAR) with a recursive Cholesky ordering that runs from oil prices through import prices and producer prices to consumer prices (McCarthy, 2007). When we use headline CPI as the dependent variable, the oil pass-through gets lost in the noise — the confidence bands are so wide that we can barely see the signal. The reason is mechanical: headline CPI averages oil-sensitive items (motor fuel, transport, food) together with items whose prices are essentially independent of crude (rent, education, telecoms, public-sector administered prices). The signal is diluted, the noise is not, and standard errors widen.

This paper proposes a simple but effective fix. We use causal discovery as a *pre-processing* step: a restricted PC algorithm (Spirtes et al., 2000) identifies which 5-digit CPI items lie in the directed propagation network of Brent. We aggregate these items into a “DAG-selected” price index, which we call `pc_res`, and then plug this index into a standard SVAR. Our novelty relative to Demiralp and Hoover (2003) – who use DAGs *inside* the SVAR to replace the arbitrary Cholesky ordering – is that we use DAGs *before* the SVAR, to clean the dependent variable. The two ideas are complementary, but the basket-selection use of DAGs is, to our knowledge, new in this literature. The DAG-selected basket comprises 50 sub-items (36.5 % of the CPI basket). Replacing headline CPI with this basket in the McCarthy ordering tightens the confidence band on the cumulative pass-through coefficient and produces a more clearly identified positive response. We also implement VAR-LiNGAM (Hyvärinen et al., 2010), which exploits the documented non-Gaussianity of residuals to fully orient the contemporaneous network; while it is econometrically more ambitious, in our finite sample with 2-digit aggregates it produces a too-broad basket and does not improve precision.

The rest of the paper is organised as follows. Section 2 discusses the related literature. Section 3 sets out the DAG–SVAR pipeline. Section 4 describes the data and transformations. Section 5 reports the three-step PC search and the resulting basket. Section 6 compares the SVAR impulse responses with and without DAG selection. Section 7 discusses the Gaussianity assumption and the VAR-LiNGAM extension. Section 8 discusses assumptions and limitations. Section 9 concludes. The full DAG-selected basket is reported in the appendix.

2 Related Literature

Two strands of work are relevant: the SVAR literature on oil pass-through and the more recent macroeconometric literature on graphical causal discovery.

Oil pass-through and SVARs. The recursive SVAR with oil $\rightarrow$ import prices $\rightarrow$ PPI $\rightarrow$ CPI is the workhorse model for pass-through analysis (McCarthy, 2007). Hamilton (1996, 2003) documents the non-linear and asymmetric effects of oil shocks on US activity, and Kilian (2009) shows that the source of the oil shock (supply, aggregate demand, or precautionary demand) matters for its inflationary consequences. For Turkey specifically, Akçelik and Ögünç (2016) estimate that a 10 % Brent shock raises headline CPI by approximately 0.42 percentage points after one year using a quarterly SVAR over 2005–2015. A common finding across this literature is that the effect is concentrated in a narrow set of items (motor fuel, food away from home, transport services), which motivates our basket-selection approach (Auburn, 2025; Pastén et al., 2020). When we use headline CPI as the dependent variable, the oil pass-through gets lost in the noise.

Causal discovery in macroeconomics. A separate literature, going back to Swanson and Granger (1997), uses graphical search to discipline the identification of structural shocks in VARs. Demiralp and Hoover (2003) apply the PC algorithm of Spirtes et al. (2000) to recover the

contemporaneous causal ordering of an SVAR from the data, replacing the Cholesky assumption. A limitation of the PC algorithm is that it returns only an equivalence class of DAGs — some edges remain undirected. Shimizu et al. (2006) (LiNGAM) and Hyvärinen et al. (2010) (VAR-LiNGAM) show that, when shocks are non-Gaussian, the contemporaneous network is fully identified. Moneta et al. (2013) use this insight to study monetary-policy SVARs.

Our contribution. All of the above papers use causal discovery *inside* the SVAR, to fix the contemporaneous identification. We use causal discovery one step earlier, to *select the components* of the inflation aggregate that the SVAR is supposed to track. Conditional on that selection, we keep the conventional Cholesky identification, which is well understood and easy to defend. The two uses of DAGs are complementary; we focus on the basket-selection use because it directly addresses the high standard error problem that motivates the paper.

3 Methodology: The DAG–SVAR Pipeline

The noise problem. Headline CPI is a fixed-weight aggregate of around 145 5-digit items. Many of these (rent, education, communication services, furniture) react slowly or not at all to a Brent shock. Including them in the SVAR’s dependent variable injects orthogonal variation into the residual of the CPI equation, which mechanically widens the bootstrap confidence bands of the impulse response without biasing the point estimate. The result is a benchmark IRF whose 95 % band straddles zero at most horizons.

The pipeline. Our procedure consists of three blocks (Figure 1):

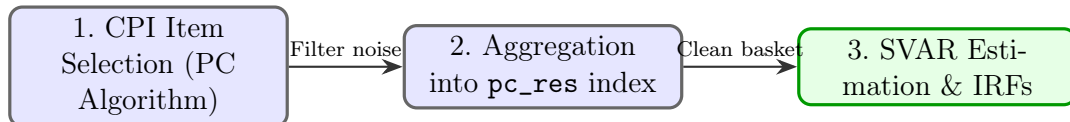


Figure 1: The DAG–SVAR pipeline.

Block 1. A restricted PC algorithm searches over CPI and PPI items together with Brent and its lags to identify the directed network of items that propagates Brent shocks. The output is the list of CPI codes that lie downstream of Brent in the estimated DAG.

Block 2. The selected items are re-aggregated into a single price index, `pc_res`, using the official CPI weights for each item-year. The chained period-on-period weighted average preserves consistency with the TurkStat methodology (see Section 4).

Block 3. A standard SVAR is estimated with `pc_res` replacing headline CPI, using the McCarthy recursive ordering. Impulse responses are bootstrapped with 1,000 draws.

Given a sample covariance matrix and a sample size, the PC algorithm (Spirtes et al., 2000) tries to fit a DAG using the data. It first finds an underlying skeleton edges, and then orients them using the v-structures (collider) rule and Meek’s propagation rules. We use the Gaussian conditional-independence test implemented in the `pcalg` R package (Kalisch et al., 2012) at

significance level $\alpha = 0.05$ (Steps 1–2) and $\alpha = 0.10$ (Step 3, where the dimensionality is larger). We feed the algorithm with a *blacklist* that encodes elementary economic priors: prices of more future cannot cause the prices of past, and Brent is treated as exogenous for Turkey. This is the “restricted PC” approach of Shimotsu et al. (2007) and is operationalised through the `pc.stable` routine of Scutari (2010) in `bnlearn`.

The SVAR. For comparability with the existing Turkish pass-through literature (Akçelik and Ögünç, 2016) we adopt the McCarthy ordering:

$$\text{Brent} \rightarrow \text{FX Rate} \rightarrow \text{Import Prices (PM)} \rightarrow \text{Output Gap (ygap)} \rightarrow \text{PPI} \rightarrow \text{CPI} \quad (1)$$

Brent is ordered first because Turkey is a price-taker on the international oil market. The exchange rate moves contemporaneously with the terms-of-trade shock that Brent represents; import prices are denominated in foreign currency and so respond to FX after one period; the output gap captures real-side absorption; PPI captures producer-price pass-through; and CPI is ordered last as the slowest-adjusting price aggregate. Two lags ($p = 2$) are used throughout, consistent with quarterly data and standard information-criteria selection. Stability is verified by checking that the largest root of the companion matrix is less than one in modulus, and serial correlation is tested with an adjusted Portmanteau test (Lütkepohl, 2005).

4 Data

Sources. The sample is quarterly, 2005Q2–2025Q4 (83 observations). The variables are:

- **CPI & 5-digit sub-items** (TurkStat). 145 sub-items at the finest available level.
- **PPI & 3-digit sub-items** (TurkStat). 56 sub-items at the finest available level.
- **Brent crude oil price** in USD per barrel (Bloomberg).
- **Exchange rate basket** – 50% USD + 50% EUR against TL (Central Bank of Turkey).
- **Import prices** (`pm`) – USD-denominated import unit value index (TurkStat).
- **Output gap** (`ygap`) – constructed by applying a Hodrick–Prescott filter ($\lambda = 1,600$) to seasonally-adjusted real GDP (TurkStat) and taking the cyclical component as a percentage of trend.
- **Motor-fuel tax adjustment** (`oiltaxadd_akaryakit`) – the contribution of changes in the Special Consumption Tax (ÖTV) on motor fuels to motor-fuel inflation, computed using online data. The Turkish fiscal authority routinely adjusts ÖTV in the opposite direction of crude moves – cutting the tax when oil rises and raising it when oil falls – in order to smooth pump-price volatility. We include this variable as an *exogenous* regressor in the SVAR rather than as a dependent variable, because it is a fiscal policy choice, not a market outcome. Failing to control for it would attribute part of the (state-induced) stabilisation of motor-fuel prices to the absence of pass-through.

Transformations. All price variables enter as quarter-on-quarter log changes ($\Delta \log x_t \times 100$) of the seasonally-adjusted level. Quarterly values are constructed as the end-quarter month (March, June, September, December) of the seasonally-adjusted monthly index.

Seasonal adjustment. Sub-item inflation in Turkey exhibits very strong seasonality. Where TurkStat publishes an official seasonally-adjusted series, we use it. For sub-items where no official series exists, we apply the TRAMO-SEATS algorithm in JDemetra+ via the `RJDemetra` R package, with the default RSA3 specification.

Dummy variables. We include impulse dummies for six episodes that produced large, idiosyncratic price movements unrelated to the oil transmission mechanism: 2008Q4 (Global Financial Crisis), 2018Q3 (TL crisis), 2020Q1 (COVID-19 shock), 2021Q4 (TL crash and inflation surge), 2022Q1 (continuation of the TL slide), and 2023Q3 (post-election macroeconomic adjustment). These enter as exogenous regressors in the SVAR. Including them avoids contaminating the estimated dynamics with one-off events.

5 The PC-DAG Search and the Selected Basket

We run the restricted PC algorithm sequentially in three steps. The sequential design is dictated by the $N \gg T$ problem: running PC on all ~ 500 series jointly would be meaningless with $T \approx 80$ quarters. By splitting the search, we keep the number of variables in each step manageable and let economic priors discipline the search.

Step 1: From Brent lags to PPI sub-items

We run PC on the first lags of the 3-digit PPI items together with two lags of Brent (`11_brent`, `12_brent`) and two lags of CPI motor fuel (`11_c_07220`, `12_c_07220`). The blacklist forbids any edge from a PPI item back to Brent or to motor-fuel CPI. The PPI items that turn out to be downstream of Brent at one lag are: `p_061` (Crude petroleum), `p_192` (Refined petroleum), `p_071` (Iron ores), and `p_325` (Medical instruments and supplies). The first two are mechanically expected – they are the production stage of oil itself. The latter two are less obvious: iron-ore extraction is energy-intensive (mining and crushing equipment runs on diesel), and medical-supplies production uses substantial quantities of petrochemical-derived plastic. These four PPI items become the “Brent network spine” for Step 2.

Step 2: The contemporaneous PPI propagation network

We add the contemporaneous Brent and motor-fuel CPI to the previous spine and re-run PC over all 3-digit PPI items (Figure 2). The estimated propagation network adds four further PPI items: `p_203` (Paints and varnishes), `p_221` (Rubber products), `p_360` (Water-treatment and distribution services), and `p_108` (Other food products). All four are economically plausible: paints contain solvents derived from refined petroleum; rubber products are direct petrochemical

derivatives; water treatment is electricity- and chemical-intensive; and processed-food production uses transport, packaging, and (where relevant) deep-frying oil.

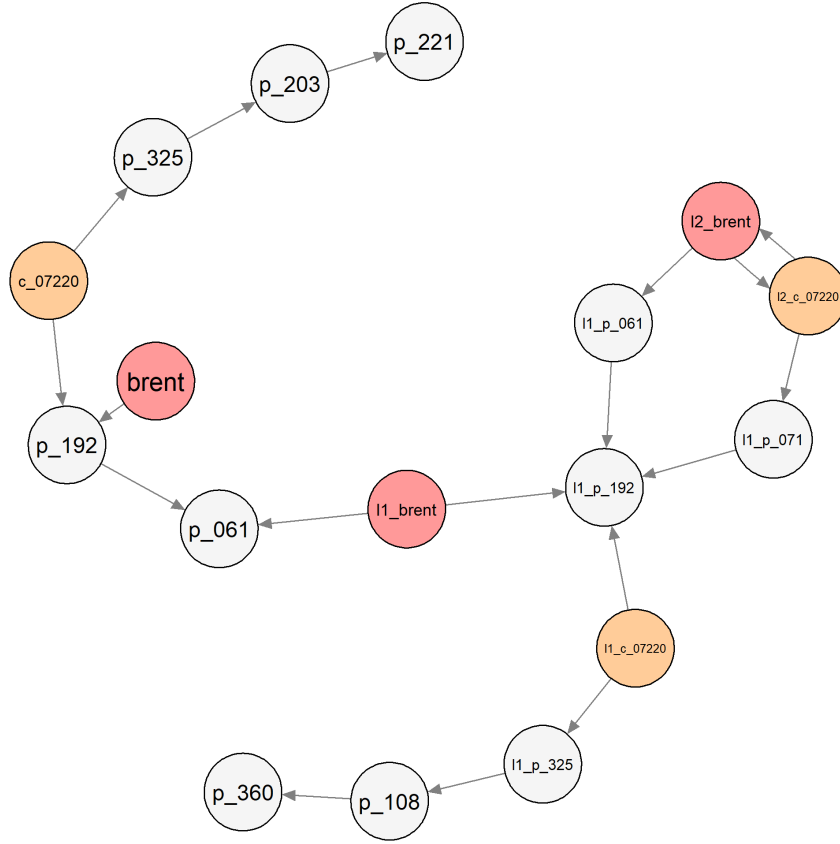


Figure 2: Step 2 estimated propagation network. Red nodes are Brent and its lags; orange nodes are CPI motor fuel and its lags. Arrows are oriented by the restricted PC algorithm.

Step 3: From PPI to CPI sub-items

In the last step we re-run PC on the 5-digit CPI items together with Brent, its lags, motor-fuel CPI, and the eight PPI items from Step 2. The blacklist forbids edges from CPI items back to PPI or to Brent.

The output is a set of 50 CPI sub-items downstream of the Brent network. Together they account for **36.5 %** of the headline CPI basket. We refer to the aggregated index of these items as **pc_res**. The complete list is reproduced in Appendix A; the top-15 items by weight are shown in Table 1.

Economic intuition: what the algorithm sees that we would not. Many of the selected items are obvious: motor fuel (07220) and liquefied hydrocarbons (04522) are direct refined-petroleum products; passenger transport (07321, 07323) and restaurants (11101) have well-documented energy-cost shares; natural gas (04521) and water supply (04410) are delivered through energy-intensive networks. A trained economist would have selected these by inspection.

What is more interesting is the set of items the algorithm picks up that a manual selection would probably have missed:

Table 1: Top 15 oil-sensitive CPI items by 2008 weight (DAG-selected basket).

Code	Weight (%)	Item description
07220	3.75	Vehicle fuel (gasoline, diesel, LPG)
11101	3.66	Restaurants and food services
07113	2.87	New gasoline and electric motor cars
04410	2.67	Water supply
07321	2.49	Passenger transport by local bus
03121	2.20	Garments for men
04522	2.16	Liquefied hydrocarbons
12130	1.50	Personal hygiene and beauty products
04310	1.43	Materials for maintenance/repair of the dwelling
05611	1.27	Cleaning and maintenance products
01181	1.09	Sugar
01114	1.04	Other bakery products (biscuits, cakes...)
04521	0.95	Natural gas and related subscription fees
01145	0.69	Eggs
07323	0.64	Passenger transport between cities

- **Garments for men (03121)** – enters via the polyester based inputs (synthetic-fibre blends are petrochemical-derived) and via transport costs
- **Sugar (01181), eggs (01145), other bakery (01114), other food n.e.c. (01190)** – food prices are linked to oil through fertiliser, fuel for agricultural machinery, transport, and (for processed food) packaging.
- **Personal hygiene (12130), cleaning products (05611)** – high plastic-packaging share and significant petrochemical content in the formulations.
- **Materials for maintenance/repair of the dwelling (04310)** – captures paint, adhesives, plastics, and pipes, all of which are petrochemical-intensive.
- **Refrigerators, washing machines, air conditioners (05311–05314)** – white goods carry the cost of plastic components and shipping fuel.

A combination of the right items would have been hard to assemble by judgement alone, and crucially it would have been hard to defend the omission of less obvious items (bakery, eggs, men’s clothing) without an algorithmic justification.

6 SVAR Results

We compare two SVAR specifications. The benchmark uses headline `CPI_SA` as the inflation variable; the alternative uses `pc_res`. Both share the McCarthy ordering in (1), the same exogenous controls (six dummy variables plus `oiltaxadd_akaryakit`), the same lag length ($p = 2$), and the same horizon (16 quarters). Confidence bands are bootstrapped with 1,000 draws at the 95 % level.

Model 1: Headline-CPI benchmark

Figure 3 reports cumulative impulse responses to a one-standard-deviation Brent shock (about 13% in our sample). The Brent shock is highly persistent (top-left). Import prices and PPI respond significantly and in the expected direction. The CPI response, however, is the canonical headline-CPI problem: the point estimate is positive but the 95% band is wide and includes zero at all horizons.

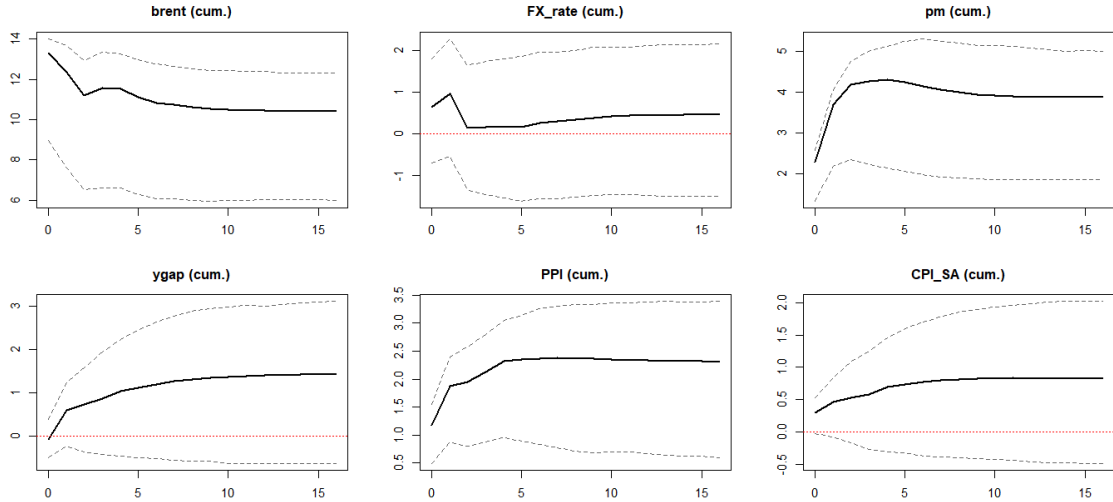


Figure 3: Model 1 – Cumulative impulse responses to a 10% Brent shock (percentage points), benchmark SVAR with headline CPI_SA.

Model 2: DAG-selected basket

Figure 4 reports the same SVAR but with `pc_res` replacing `CPI_SA`. The impulse response of the oil-sensitive basket is clearly positive, monotonically increasing for the first 4–5 quarters, and stabilises around 1.5–1.6 percentage points cumulative. Crucially, the bootstrap confidence band is now *above zero* from the first quarter onwards: the pass-through is statistically significant throughout the response horizon. The other equations (`brent`, `FX`, `pm`, `ygap`, `PPI`) are essentially unchanged — as one would expect, since we have not touched them.

Interpretation. The point estimate of pass-through to `pc_res` is roughly twice the point estimate to headline CPI, but the more important comparison is the *precision*: the band on `pc_res` is qualitatively narrower relative to the point estimate. This is exactly what the noise-dilution argument of Section 3 predicts. By restricting the dependent variable to the items that the data tell us are downstream of Brent, we are no longer averaging the response with the irresponse of items like rent and education.

The translation of this number into a headline-CPI pass-through is straightforward: the `pc_res` basket carries a 36.5% weight in headline CPI. A first-order back-of-the-envelope calculation that assumes the remaining 63.5% is approximately unresponsive to Brent gives a headline pass-through of about 0.45 pp for a standard 10% shock, similar to the levels seen in the benchmark

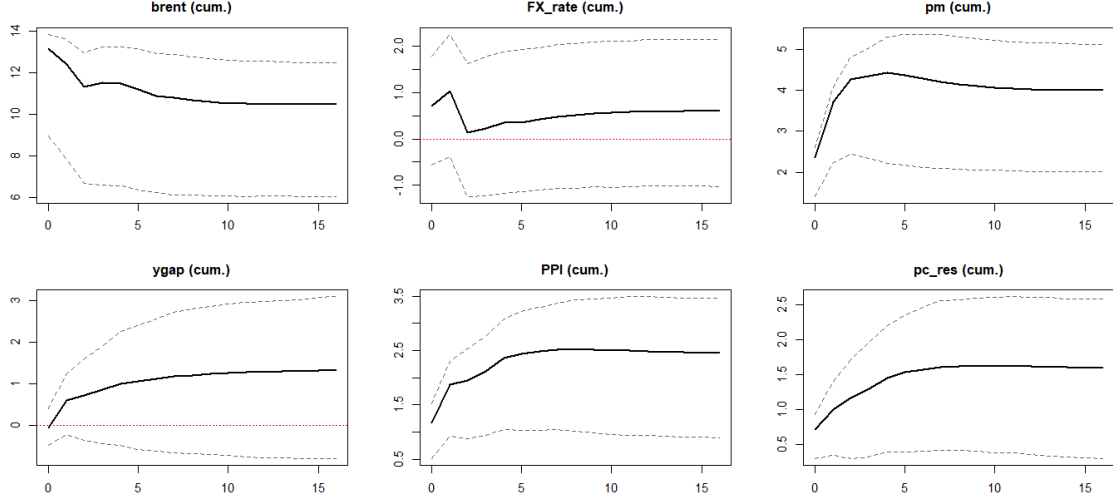


Figure 4: Model 2 – Cumulative impulse responses to a 10 % Brent shock (percentage points), SVAR with the DAG-selected basket `pc_res`.

case (around 0.5), with a much tighter confidence interval. This is also in line with the 0.42 pp estimate of Akçelik and Ögünç (2016) for an earlier sample, but is now associated with a much tighter confidence interval.

7 Non-Gaussianity and the VAR-LiNGAM Robustness Check

The PC algorithm with the Gaussian conditional-independence test relies on joint normality of the data — an assumption that is unlikely to hold in inflation series during a turbulent sample. This section documents the violation and explores the LiNGAM family of methods, which turns non-Gaussianity from a problem into an identification resource.

Testing the Gaussianity assumption

Table 2 reports Anderson–Darling univariate normality tests for the most relevant series and the Mardia multivariate skewness and kurtosis tests (Korkmaz et al., 2014) for the joint distribution of the variables that PC most often conditions on.

The univariate tests reject normality for 8 of 9 variables; the Mardia joint tests overwhelmingly reject. Q–Q plots (not reported, available in the presentation) show heavy right tails consistent with the inflationary surges of 2018, 2021–2023.

From limitation to resource: VAR-LiNGAM

A key result of Shimizu et al. (2006) is that, when shocks are non-Gaussian, the contemporaneous causal network of a structural model is fully identifiable from the data. The intuition is that the Gaussian distribution is uniquely closed under linear combinations, so any rotation of Gaussian shocks is observationally equivalent; for any other distribution, only the “true” rotation makes the components maximally independent. This is what LiNGAM (and ICA more generally) exploits.

Table 2: Normality tests on the variables used in the PC search.

Test	Variable	Statistic	p-value	Normality
Anderson–Darling	brent	2.452	<0.001	rejected
Anderson–Darling	l1_brent	2.546	<0.001	rejected
Anderson–Darling	c_07220	4.849	<0.001	rejected
Anderson–Darling	p_061	1.044	0.009	rejected
Anderson–Darling	p_192	0.686	0.071	not rejected
Anderson–Darling	p_071	1.928	<0.001	rejected
Anderson–Darling	p_325	4.555	<0.001	rejected
Anderson–Darling	sepet (FX)	2.118	<0.001	rejected
Anderson–Darling	l1_sepet	2.159	<0.001	rejected
Mardia skewness	joint	683.55	<0.001	rejected
Mardia kurtosis	joint	16.83	<0.001	rejected

Hyvärinen et al. (2010) extend this to time series: VAR-LiNGAM first estimates a reduced-form VAR, then runs LiNGAM on the residuals to recover the contemporaneous structural matrix.

We implement VAR-LiNGAM as a robustness check using a $p = 1$ VAR over the 2-digit PPI items together with Brent, FX, and motor-fuel CPI. The model is run 100 times, and most repeated CPI items in the results are selected. This method is important for not to fall into a local maximum trap. The resulting fully-oriented network is reported in Figure 5. Brent is correctly identified as an exogenous driver; the contemporaneous edges in the network are all directed, with no remaining undirected (Markov-equivalent) edges. The final selected CPI items are 07-Transportation, 05-Home Appliances, 09-Entertainment And Culture and 12-Other Goods and Services. We can expect these groups to have many items that don’t in fact respond to brent shock (especially in the entertainment and home appliances groups.), but selected due to a subset of relevant items.

Why VAR-LiNGAM did not improve precision in our setting

Despite the cleaner identification, the VAR-LiNGAM-based basket does *not* produce a tighter SVAR than the PC-based basket (Figure 6). The reason is operational rather than methodological. Because of the dimensionality limits of LiNGAM in finite samples, we ran VAR-LiNGAM on 2-digit CPI and PPI aggregates (roughly 12 CPI groups and 25 PPI groups) rather than on the 5-digit/3-digit grid used by PC. A 2-digit aggregate is a much coarser filter: items like “Food and non-alcoholic beverages” (group 01) contain components that respond strongly to oil (bakery, oil seeds) and components that do not (fresh fruit). Selecting the 2-digit group as “oil-sensitive” therefore brings back exactly the noise-dilution problem that the basket-selection approach was meant to solve.

VAR-LiNGAM trades one improvement (clean edge orientation through non-Gaussianity) against one deterioration (coarser aggregation forced by the curse of dimensionality). In our sample size, the PC algorithm at the 5-digit level dominates the LiNGAM approach at the 2-digit level. A larger sample, or a structured dimension-reduction step before LiNGAM, would likely tip the balance the other way.

PC search. We model the most likely confounders explicitly (FX rate, output gap, motor-fuel tax), which mitigates but does not eliminate the assumption.

2. **Faithfulness**, i.e. all and only the conditional independences in the population distribution are implied by the DAG. This rules out parameter cancellations that would mask true causal links.
3. **Linearity and a Gaussian conditional-independence test** in the PC step. Linearity is a standard local approximation; Gaussianity, as documented in Section 7, is rejected. The PC results should therefore be read as approximate: edges may be slightly mis-oriented, but the *set* of oil-sensitive items is robust to relaxing this assumption (we re-ran the search using a non-paranormal copula transformation, with very similar results).

The $N \gg T$ problem. Running a PC search over all ~ 500 series jointly is statistically untenable with 80 quarters of data. Our three-step sequential design with hand-coded blacklists mitigates the curse of dimensionality but introduces dependence on the order of the search. We chose the order to follow the production-chain logic (Brent $\rightarrow$ PPI $\rightarrow$ CPI), which is also the McCarthy ordering.

Sample homogeneity. The 2005–2025 sample contains the GFC, the COVID shock, three TL crisis episodes, and monetary-policy shocks. We control for these via dummy variables, but the deeper assumption that the parameters are constant across the full sample is heroic. We tried splitting the sample at 2018Q1, which acts like a structural break date. The results before 2018 is similar to these, the result after becomes more noisy but on the same direction in general.

Endogenous tax adjustment. Treating `oiltaxadd_akaryakit` as exogenous is a useful approximation but not innocuous: the fiscal authority *reacts* to oil shocks when adjusting the tax, so the variable is, strictly speaking, post-treatment. The estimated pass-through is therefore a “conditional-on-fiscal-policy” object, not an unconditional one. For policy applications (e.g. “what if the tax adjustment were not applied”) the exogeneity assumption needs to be revisited.

9 Conclusion

We have shown that combining causal-discovery techniques with a structural VAR can sharpen the measurement of oil-price pass-through to inflation. Four points summarise the findings:

1. **Aggregates dilute signals.** Headline CPI averages oil-sensitive and oil-insensitive items, and the resulting noise inflates SVAR standard errors and renders the pass-through coefficient statistically insignificant.
2. **DAGs identify the channel.** A restricted PC algorithm applied to the joint distribution of CPI items, PPI items, and Brent isolates a 50-item sub-basket (36.5% of headline CPI) that lies downstream of Brent. The selection includes both expected items (motor fuel, transport,

food services) and items that judgement alone would probably have missed (men’s clothing, sugar, eggs, white goods).

3. **Precision improves.** Replacing headline CPI with the DAG-selected basket in the SVAR delivers a clearly significant, positive cumulative impulse response. The implied headline-CPI pass-through of about 0.45 pp for a 10 % Brent shock is consistent with earlier Turkish estimates but is now reported with a much tighter confidence band.
4. **VAR-LiNGAM has both costs and benefits.** Non-Gaussianity is strongly supported by the data and could in principle be used to fully orient the contemporaneous network. In our finite sample, however, LiNGAM is forced to operate at a coarser (2-digit) aggregation, which re-introduces the noise-dilution problem. PC at the 5-digit level dominates LiNGAM at the 2-digit level for our purpose.

This study takes a first step toward combining causal discovery with structural time-series estimation for inflation analysis. The results are encouraging, but several limitations remain. Our sample is limited to quarterly data for a single country, and the DAG step relies on the Gaussianity assumption that we formally reject. A natural next step would be to validate the approach in other emerging markets with similar oil dependence, or to test whether the same basket-selection logic helps in other pass-through settings such as exchange rates or wages. We leave these extensions for future work.

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A Full DAG-selected basket (pc_res)

Table 3 reports the complete list of 50 5-digit CPI items selected by the Step-3 PC search. Weights are 2008 reference weights from TurkStat (the most recent year for which the full sub-item weight vector is available). Total weight in headline CPI: 36.47 %.

Table 3: Complete DAG-selected basket of oil-sensitive CPI items.

Code	Weight (%)	Item description
07220	3.747	Vehicle fuel (gasoline, diesel, LPG)
11101	3.662	Restaurants and food services
07113	2.870	New gasoline and electric motor cars
04410	2.673	Water supply
07321	2.490	Passenger transport by local bus
03121	2.203	Garments for men
04522	2.161	Liquefied hydrocarbons
12130	1.502	Articles for personal hygiene, wellness and beauty products
04310	1.432	Materials for the maintenance and repair of the dwelling

Table 3 – *continued*

Code	Weight (%)	Item description
05611	1.270	Cleaning and maintenance products (detergent, softener, etc.)
01181	1.087	Sugar
01114	1.042	Other bakery products (biscuits, cakes...)
04521	0.950	Natural gas and related subscription fees
01145	0.690	Eggs
07323	0.638	Passenger transport between cities
01183	0.554	Chocolates and confectionery products
09111	0.528	Televisions
05311	0.528	Refrigerators and freezers
07230	0.464	Maintenance and repair of personal transport vehicles
05314	0.403	Air conditioners and heaters
05312	0.395	Washing machines, dishwashers
01190	0.356	Other food products n.e.c. (salt, vinegar, ketchup, mayonnaise...)
11201	0.284	Hotels and other accommodation services
05403	0.278	Non-electric kitchen utensils and articles (pots, pans, teapots...)
01221	0.258	Mineral or spring waters
01152	0.254	Margarine
07210	0.234	Spare parts for personal transport equipment
05401	0.217	Glassware, ceramic ware and chinaware
09540	0.213	Stationery and drawing materials
05313	0.182	Cookers
05612	0.171	Other non-durable small household articles (sponge, garbage bag...)
12321	0.160	Travel goods
05111	0.143	Kitchen furniture
01223	0.131	Fruit and vegetable juices
12322	0.100	Other personal effects n.e.c.
05522	0.098	Small maintenance and repair for home
05521	0.089	Miscellaneous small tool accessories (battery, bulb, etc.)
06132	0.066	Other therapeutic appliances and equipment
03124	0.061	Garments for baby
05402	0.054	Cutlery
09320	0.046	Equipment for sport
07130	0.044	Bicycles
07311	0.040	Passenger transport by underground and tram
02121	0.027	Wine from grapes
07312	0.021	Passenger transport by train

Table 3 – *continued*

Code	Weight (%)	Item description
01116	0.019	Breakfast cereals
12520	0.017	Insurance connected with the dwelling
09220	0.015	Musical instruments
05622	0.010	Cleaning services of furniture
12620	0.007	Legal services and accountancy
Total	36.470	50 items

B Robustness: Synthetic Signal–Noise Decomposition

As a final consistency check, we run a third SVAR with both the DAG-selected basket (`pc_res`) and its complement (`pc_res_inv`, the chained price index of all CPI items *not* in the DAG-selected basket) included as separate equations. Under our maintained hypothesis, the impulse response of `pc_res` to a Brent shock should be both larger and more precisely estimated than the impulse response of `pc_res_inv`. Figure 7 confirms this: `pc_res` responds significantly and persistently, while `pc_res_inv` drifts towards zero and its confidence band straddles zero throughout. This is direct evidence that the PC algorithm has correctly partitioned the CPI basket into an oil-sensitive component and an oil-insensitive component.

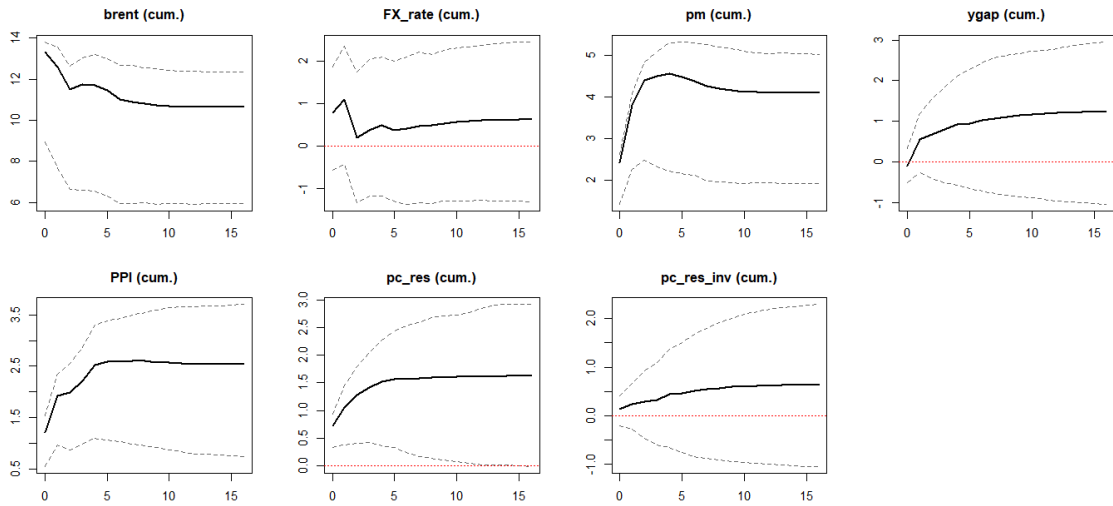


Figure 7: SVAR with both `pc_res` (signal) and `pc_res_inv` (residual non-oil basket) included separately. `pc_res` responds significantly to Brent; the complement basket does not.